C Language: IIITB COMET FOUNDATION

1. Explain pointers in C. Write a program to demonstrate pointer arithmetic and pointer to pointer.
2. Describe memory management in C. How does malloc differ from calloc? Write code to illustrate both.
3. Write a function to reverse a linked list using pointers. Explain its time and space complexity.
4. What are the differences between structure and union in C? Provide examples showcasing memory usage differences.
5. Write a C program to implement a circular queue using arrays and explain how overflow and underflow conditions are handled.
6. Explain the use of type def in C. How can it be used to simplify complex declarations?
7. Write a program to find the factorial of a number using both recursion and iteration. Compare their stack usage.
8. What are static and dynamic memory? Write an example of a program using static and dynamic variable allocation.
9. Explain the difference between call by value and call by reference with suitable program examples.
10. Write a C program to demonstrate dangling pointers and how to avoid them.
11. Explain the significance of const qualifier in pointer declarations with examples.
12. Describe the importance of header files and how #include works in C.
13. Write a program to implement string manipulation functions such as strlen, strcat, and strcmp without using the standard library.
14. What are function pointers? Write a program that demonstrates using function pointers to select and invoke different functions dynamically.
15. How are arrays passed to functions in C? Write a program to sort an array using function pointers.
16. Describe the concept of bit-fields in structures. Write a program to demonstrate their usage.
17. Explain volatile keyword in C. When would you use it? Provide an example.
18. Write a program to detect memory leaks in dynamically allocated memory.
19. Explain how the preprocessor works in C. Write macros for finding maximum and minimum of two numbers.
20. What is the difference between volatile and const keyword? Explain with examples.
21. Write a C program to implement a simple calculator using switch-case statements.
22. Define recursion. Write a C program to calculate Fibonacci numbers using recursion and memoization.
23. What are dangling pointers and wild pointers? How to avoid them in C?
24. Explain alignment and padding in structures. How can you minimize padding?

25. Write a program to simulate a stack using linked lists in C.
26. Describe segmentation fault with examples and how to debug it.
27. Write a program to parse command line arguments and process options.
28. Explain the use of the register storage class. Is it still relevant in modern compilers?
29. Write C programs to demonstrate implicit type conversion and explicit casting.
30. What is undefined behavior in C? Give examples from common mistakes.
31. Explain the difference between struct and type def struct with examples.
32. Write a program to merge two sorted linked lists into a single sorted linked list.
33. What are the differences between malloc, calloc, realloc, and free? Write code snippets for each.
34. Explain the role of static keyword in variables, functions, and global scope.
35. Write a program to count the frequency of each character in a string using arrays.
36. How does the switch statement work with enumeration types in C? Provide example.
37. Write a program to dynamically allocate a 2D array using malloc.
38. Explain the use of the sentinel node concept in linked list implementation.
39. Write a C program to implement a binary search tree with insertion and traversal.
40. Describe the difference between shallow copy and deep copy. Write a program to do deep copy of a structure containing pointers.
41. Explain the memory layout of a C program (Stack, Heap, Data, Text segments).
42. Write a program to detect palindrome strings using pointers.
43. Describe how C implements pass-by-reference using pointers with examples.
44. Write a function to swap two numbers using pointers without using a temporary variable.
45. Explain storage classes in C with examples.
46. Write a program to handle file I/O operations including reading, writing, and appending text files.
47. Describe how to implement linked lists with dynamic memory allocation in C.
48. Write a program to demonstrate the use of break and continue statements in loops.
49. How can recursion lead to stack overflow? Write an example where infinite recursion occurs.
50. Write a program in C to implement a priority queue using arrays.